

10. Intersections with planes

In the last chapter we saw that if A is a Borel set in $\mathbf{R}^n$ with $\dim A = s > m$, then for $\gamma_{n,m}$ almost all $V \in G(n, m)$, $\mathcal{H}^m(P_V A) > 0$. This means that the set of those $a \in V$ for which $A \cap P_V^{-1}\{a\} \neq \emptyset$ has positive $\mathcal{H}^m$ measure for $\gamma_{n,m}$ almost all $V \in G(n, m)$. Here we shall say much more; “generically” the $(n - m)$ -planes $P_V^{-1}\{a\}$ intersect A in a set of dimension $s - m$.

The upper bound $s - m$ for the dimension of the intersections follows easily from the general Fubini-type inequality for Hausdorff measures and Lipschitz maps proved in Theorem 7.7. It is the lower bound that is harder to obtain. For this we shall again use capacities. Given A with $C_s(A) > 0$, we shall consider a Radon measure $\mu \in \mathcal{M}(A)$ with $I_s(\mu) < \infty$. The main problem will be to construct non-zero Radon measures in $A \cap P_V^{-1}\{a\}$ with finite $(s - m)$ -energy for many intersections $A \cap P_V^{-1}\{a\}$. We shall do this by a disintegration applying the differentiation theory of Chapter 2 to $P_{V\#}\mu$.

Applying Theorem 7.7 to P_V , $V \in G(n, m)$, we see that if $A \subset \mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$, $m \leq s \leq n$, then for $\mathcal{H}^m$ almost all $a \in V$, $\mathcal{H}^{s-m}(A \cap P_V^{-1}\{a\}) < \infty$, whence $\dim A \cap P_V^{-1}\{a\} \leq s - m$. We shall now pursue the lower bound.

Slicing measures with planes

10.1. First we shall “slice” an arbitrary Radon measure on $\mathbf{R}^n$ with parallel $(n - m)$ -planes. Fix $V \in G(n, m)$ and let $W = V^\perp \in G(n, n - m)$. Then

$$P_V^{-1}\{a\} = W + a \equiv W_a \quad \text{for } a \in V.$$

Denote by $C_0^+(\mathbf{R}^n)$ the space of continuous non-negative functions on $\mathbf{R}^n$ with compact support. Equipped with a distance d ,

$$d(\varphi, \psi) = \sup \{ |\varphi(x) - \psi(x)| : x \in \mathbf{R}^n \},$$

it is a separable metric space (recall the proof of Theorem 1.23).

Let μ be a Radon measure on $\mathbf{R}^n$. For each $\varphi \in C_0^+(\mathbf{R}^n)$ we define a Radon measure ν_φ setting

$$\nu_\varphi(A) = \int_A \varphi \, d\mu$$

for Borel sets $A \subset \mathbf{R}^n$. Then $P_{V\sharp}\nu_\varphi$ is a Radon measure by Theorem 1.18 and the differentiation theorem 2.12 (1) guarantees the existence of the finite limit

$$(10.2) \quad \begin{aligned} \mu_{W,a}(\varphi) &= \lim_{\delta \downarrow 0} (2\delta)^{-m} P_{V\sharp}\nu_\varphi(B(a, \delta)) \\ &= \lim_{\delta \downarrow 0} (2\delta)^{-m} \int_{W_a(\delta)} \varphi \, d\mu \end{aligned}$$

for $\mathcal{H}^m$ almost all $a \in V$ (recall that $E(\delta) = \{x : d(x, E) \leq \delta\}$). We would like, for $\mathcal{H}^m$ almost all $a \in V$, to identify $\mu_{W,a}$ with a Radon measure on W_a in such a way that $\mu_{W,a}(\varphi)$ would equal the integral $\int \varphi \, d\mu_{W,a}$. By the Riesz representation theorem 1.16 we can do this as soon as we know that $\mu_{W,a}$ satisfies

$$\mu_{W,a}(\alpha\varphi + \beta\psi) = \alpha\mu_{W,a}(\varphi) + \beta\mu_{W,a}(\psi)$$

for $\alpha, \beta > 0$ and $\varphi, \psi \in C_0^+(\mathbf{R}^n)$. (Then $\mu_{W,a}$ extends to a positive linear functional on $C_0(\mathbf{R}^n)$.) The only problem is that we need to know that for $\mathcal{H}^m$ almost all $a \in V$, $\mu_{W,a}(\varphi)$ is defined by (10.2) for all $\varphi \in C_0^+(\mathbf{R}^n)$, that is, the exceptional set of the points a is independent of φ . This is a consequence of the separability of $C_0^+(\mathbf{R}^n)$. In fact, let D be a countable dense subset of $C_0^+(\mathbf{R}^n)$, for $\varphi \in D$ let E_φ be the set of those $a \in V$ for which the finite limit in (10.2) fails to exist, and let $E = \bigcup_{\varphi \in D} E_\varphi$. Then $\mathcal{H}^m(E) = 0$, and it is straightforward to see that for $a \in V \setminus E$ the limit in (10.2) exists and is finite for all $\varphi \in C_0^+(\mathbf{R}^n)$. Thus we can conclude that for $\mathcal{H}^m$ almost all $a \in V$ there exists a Radon measure $\mu_{W,a}$ such that for all $\varphi \in C_0^+(\mathbf{R}^n)$,

$$(10.3) \quad \int \varphi \, d\mu_{W,a} = \lim_{\delta \downarrow 0} (2\delta)^{-m} \int_{W_a(\delta)} \varphi \, d\mu.$$

We shall now examine some further properties of these measures. First, (10.3) gives immediately

$$(10.4) \quad \text{spt } \mu_{W,a} \subset W_a \cap \text{spt } \mu.$$

Let $g: \mathbf{R}^n \rightarrow [0, \infty]$ be lower semicontinuous. Then g is a limit of a non-decreasing sequence $\varphi_i \in C_0^+(\mathbf{R}^n)$. Thus (10.3) implies

$$(10.5) \quad \int g \, d\mu_{W,a} \leq \liminf_{\delta \downarrow 0} (2\delta)^{-m} \int_{W_a(\delta)} g \, d\mu.$$

Recalling the interpretation of $\int \varphi d\mu_{W,a}$ as the derivative of $P_{V\sharp}\nu_\varphi$ in (10.2) and using Theorem 2.12 (2), we obtain for any Borel set $B \subset V$,

$$(10.6) \quad \int_B \int \varphi d\mu_{W,a} d\mathcal{H}^m a \leq \int_{P_V^{-1}(B)} \varphi d\mu$$

with equality if $P_{V\sharp}\mu \ll \mathcal{H}^m$. This readily extends to lower semicontinuous functions.

In particular, if $P_{V\sharp}\mu \ll \mathcal{H}^m$, then $\mu(\mathbf{R}^n) = \int \mu_{W,a}(\mathbf{R}^n) d\mathcal{H}^m a$ so that many of the measures $\mu_{W,a}$ are not zero if μ is not. Heading towards capacities, we shall now show that many of them have finite $(s-m)$ -energy provided μ has finite s -energy and $s > m$.

10.7. Theorem. *There is a constant c depending only on n and m such that for $m < s < n$ and any Radon measure μ on $\mathbf{R}^n$,*

$$\iint_{W^\perp} I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a d\gamma_{n,n-m} W \leq cI_s(\mu).$$

Proof. We skip the rather simple measurability arguments; the reader may consult Mattila [4]. Notice that the function $x \mapsto |x|^{m-s}$ is lower semicontinuous. Thus applying (10.5), Fatou's lemma, and Fubini's theorem twice, we obtain

$$\begin{aligned} & \int_{W^\perp} \int I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a d\gamma_{n,n-m} W \\ & \leq \liminf_{\delta \downarrow 0} (2\delta)^{-m} \int_{W^\perp} \int_{W_a(\delta)} \int |x-y|^{m-s} d\mu x d\mu_{W,a} y d\mathcal{H}^m a d\gamma_{n,n-m} W \\ & = \liminf_{\delta \downarrow 0} (2\delta)^{-m} \int_{W^\perp} \int_{W_a(\delta)} \int |x-y|^{m-s} d\mu_{W,a} y d\mu x d\mathcal{H}^m a d\gamma_{n,n-m} W \\ & = \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint \int_{\{a \in W^\perp : d(x, W_a) \leq \delta\}} \int |x-y|^{m-s} d\mu_{W,a} y d\mathcal{H}^m a d\mu x \\ & \quad \times d\gamma_{n,n-m} W. \end{aligned}$$

To the two innermost integrals we can apply the inequality (10.6) with $B = \{a \in W^\perp : d(x, W_a) \leq \delta\}$, whence $P_{W^\perp}^{-1}(B) = \{y : |P_{W^\perp}(x-y)| \leq$

$\delta\}$. Using also Fubini's theorem, (3.10) and Lemma 3.11, we get

$$\begin{aligned}
 & \int_{W^\perp} \int I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a d\gamma_{n,n-m} W \\
 & \leq \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint \int_{\{y: |P_{W^\perp}(x-y)| \leq \delta\}} |x-y|^{m-s} d\mu y d\mu x d\gamma_{n,n-m} W \\
 & = \liminf_{\delta \downarrow 0} (2\delta)^{-m} \iint |x-y|^{m-s} \gamma_{n,n-m}(\{W : |P_{W^\perp}(x-y)| \leq \delta\}) d\mu y \\
 & \quad \times d\mu x \\
 & \leq cI_s(\mu)
 \end{aligned}$$

as required. $\square$

Plane sections, capacities and Hausdorff measures

We are now ready to handle the capacities.

10.8. Theorem. *There is a constant c depending only on n and m such that for $m < s < n$ and $A \subset \mathbf{R}^n$,*

$$\int^* \left(\int_{*W^\perp} C_{s-m}(A \cap W_a) d\mathcal{H}^m a \right)^{-1} d\gamma_{n,n-m} W \leq cC_s(A)^{-1}.$$

In particular, if $C_s(A) > 0$, then for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$,

$$\mathcal{H}^m(\{a \in W^\perp : C_{s-m}(A \cap W_a) > 0\}) > 0.$$

Proof. We may assume $C_s(A) > 0$. Let $\varepsilon > 0$ and let μ be a Radon measure with compact support such that $\text{spt } \mu \subset A$, $\mu(A) = 1$ and $I_s(\mu) \leq C_s(A)^{-1} + \varepsilon$. Then $I_s(\mu) < \infty$ and, as $m < s$ and $\text{spt } \mu$ is compact, also $I_m(\mu) < \infty$. Therefore by Theorem 9.7 and (3.10), $P_{W^\perp \#} \mu \ll \mathcal{H}^m$ for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$. Thus by (10.6) for any such W ,

$$(1) \quad \int_{W^\perp} \mu_{W,a}(\mathbf{R}^n) d\mathcal{H}^m a = \mu(\mathbf{R}^n) = 1.$$

Let

$$E_W = \{a \in W^\perp : \mu_{W,a}(\mathbf{R}^n) > 0\}$$

and set

$$\nu_{W,a} = \mu_{W,a}(\mathbf{R}^n)^{-1} \mu_{W,a} \quad \text{for } a \in E_W.$$

Then by (10.4) $\text{spt } \nu_{W,a} \subset \text{spt } \mu \cap W_a \subset A \cap W_a$ and $\nu_{W,a}(A \cap W_a) = 1$, whence $I_{s-m}(\nu_{W,a})^{-1} \leq C_{s-m}(A \cap W_a)$. From Theorem 10.7 we see that for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$, $I_{s-m}(\nu_{W,a}) < \infty$ for $\mathcal{H}^m$ almost all $a \in E_W$. Hence by (1) and Hölder's inequality we have for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$,

$$\begin{aligned} 1 &= \left(\int_{W^\perp} \mu_{W,a}(\mathbf{R}^n) d\mathcal{H}^m a \right)^2 \\ &= \left(\int_{E_W} \mu_{W,a}(\mathbf{R}^n) I_{s-m}(\nu_{W,a})^{1/2} I_{s-m}(\nu_{W,a})^{-1/2} d\mathcal{H}^m a \right)^2 \\ &\leq \int_{E_W} \mu_{W,a}(\mathbf{R}^n)^2 I_{s-m}(\nu_{W,a}) d\mathcal{H}^m a \int_{E_W} I_{s-m}(\nu_{W,a})^{-1} d\mathcal{H}^m a \\ &= \int_{W^\perp} I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a \int_{E_W} I_{s-m}(\nu_{W,a})^{-1} d\mathcal{H}^m a \\ &\leq \int_{W^\perp} I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a \int_{*W^\perp} C_{s-m}(A \cap W_a) d\mathcal{H}^m a. \end{aligned}$$

Integrating over $G(n, n-m)$ we have by Theorem 10.7

$$\begin{aligned} &\int^* \left(\int_{*W^\perp} C_{s-m}(A \cap W_a) d\mathcal{H}^m a \right)^{-1} d\gamma_{n,n-m} W \\ &\leq \int \int_{W^\perp} I_{s-m}(\mu_{W,a}) d\mathcal{H}^m a d\gamma_{n,n-m} W \\ &\leq c I_s(\mu) \leq c(C_s(A)^{-1} + \varepsilon). \end{aligned}$$

Letting $\varepsilon \downarrow 0$ we obtain the desired inequality. $\square$

10.9. Remark. Again, as in the case of projections in 9.5, Theorem 10.8 leads to an inequality for the integral of $C_{s-m}(A \cap W_a)$ in place of its reciprocal. Also the opposite inequality holds. This follows from a capacity analogue of Theorem 7.7 proved in Mattila [6], see also Sadullaev [1]. Thus

$$\begin{aligned} c(n, m) C_s(A) &\leq \int^* \int_{*W^\perp} C_{s-m}(A \cap W_a) d\mathcal{H}^m a d\gamma_{n,n-m} W \\ &\leq c(n, m, s) C_s(A) \end{aligned}$$

for $A \subset \mathbf{R}^n$ and $m < s < n$. In fact, the right hand inequality holds also for $s = m$ provided C_0 is defined as the logarithmic capacity, see Mattila [6] and Sadullaev [1].

We now use the measures $\mu_{W,a}$ to derive information about the Hausdorff dimension of the $(n-m)$ -plane sections.

10.10. Theorem. Let $m \leq t \leq n$ and let $A \subset \mathbf{R}^n$ be a Borel set with $0 < \mathcal{H}^t(A) < \infty$. Then for all $W \in G(n, n-m)$,

$$\mathcal{H}^{t-m}(A \cap W_a) < \infty \quad \text{for } \mathcal{H}^m \text{ almost all } a \in W^\perp,$$

and for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$,

$$\mathcal{H}^m(\{a \in W^\perp : \dim(A \cap W_a) = t-m\}) > 0.$$

Proof. The first assertion was already observed at the beginning of this chapter as an immediate consequence of Theorem 7.7. For the second we have to show that for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$, $\dim(A \cap W_a) \geq t-m$ for $a \in W^\perp$ in a set of positive $\mathcal{H}^m$ measure. We may assume $m < t$. By Frostman's lemma 8.8 there is $\mu \in \mathcal{M}(A)$ such that $\mu(B(x, r)) \leq r^t$ for $x \in \mathbf{R}^n$, $r > 0$. As noted at the beginning of Chapter 8, $I_s(\mu) < \infty$ for $s < t$. In particular, $I_m(\mu) < \infty$, and by Theorem 9.7, $P_{W^\perp \#} \mu \ll \mathcal{H}^m$ for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$. For such a W let

$$E_W = \{a \in W^\perp : \mu_{W,a}(\mathbf{R}^n) > 0\}.$$

Let $m < s < t$. As $I_s(\mu) < \infty$, Theorem 10.7 gives for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$, $I_{s-m}(\mu_{W,a}) < \infty$ for $\mathcal{H}^m$ almost all $a \in W^\perp$. Since $\text{spt } \mu_{W,a} \subset A \cap W_a$, this implies $C_{s-m}(A \cap W_a) > 0$ for $\mathcal{H}^m$ almost all $a \in E_W$, whence by Theorem 8.9, $\dim A \cap W_a \geq s-m$. This being true for all $m < s < t$, we have for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$ that $\dim A \cap W_a \geq t-m$ for $\mathcal{H}^m$ almost all $a \in E_W$ (use a sequence $s_i \uparrow t$ to find exceptional subsets of $G(n, n-m)$ and E_W independent of s). It remains to verify that $\mathcal{H}^m(E_W) > 0$ for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$. But this holds by the formula (1) in the proof of Theorem 10.8 whenever $P_{W^\perp \#} \mu \ll \mathcal{H}^m$; in particular for $\gamma_{n,n-m}$ almost all $W \in G(n, n-m)$. $\square$

Using the same ingredients one can proceed to other results of a similar kind, see Mattila [4]. We give one of them without proof.

10.11. Theorem. If $m < s < n$ and A is an $\mathcal{H}^s$ measurable subset of $\mathbf{R}^n$ with $\mathcal{H}^s(A) < \infty$, then

$$\dim(A \cap (W + x)) = s-m \quad \text{and} \quad \mathcal{H}^{s-m}(A \cap (W + x)) < \infty$$

for $\mathcal{H}^s \times \gamma_{n,n-m}$ almost all $(x, W) \in A \times G(n, n-m)$.

10.12. Remarks. Theorems 10.10 and 10.11 were first proved by Marstrand [1] in the case $m = 1$, $n = 2$, and later generalized in

Mattila [2]. The above method is from Mattila [4]. Marstrand also constructed an example of a compact set A in $\mathbf{R}^2$ showing that it can happen that $\mathcal{H}^{s-1}(A \cap (W+x)) = 0$ for $\mathcal{H}^s \times \gamma_{2,1}$ almost all $(x, W) \in A \times G(2, 1)$ although $0 < \mathcal{H}^s(A) < \infty$.

It is not known whether the second statement of Theorem 10.11 holds for $s = m$; the first one obviously does. For example in $\mathbf{R}^2$ the problem is: does $\mathcal{H}^1(A) < \infty$ imply that for $\mathcal{H}^1$ almost all $x \in A$ almost all lines through x meet A in a finite set? See also the remark (1) in 18.10.

The methods of this chapter can be modified to give similar results when the $(n-m)$ -planes are replaced by isometric images of some fixed $(n-m)$ -dimensional C^1 submanifold of $\mathbf{R}^n$, see Mattila [7]. In Chapter 13 we shall study the more general problem of determining the generic dimension of the intersections of two arbitrary Borel sets moving in $\mathbf{R}^n$. For an application of the results of this chapter, see Lang [1].

Benjamini and Peres [1] studied, see also Kenyon and Peres [1], the dimension of the intersections of E with lines in a fixed direction for some self-similar sets $E \subset \mathbf{R}^2$. Davies and Fast [1] gave an elegant method to construct Borel sets E of Hausdorff dimension n in $\mathbf{R}^n$ such that for uncountably many lines $L \in G(n, 1)$, $(L+a) \cap E$ contains at most one point for all $a \in \mathbf{R}^n$. Fast [1] showed that if two rectifiable plane curves intersect every line in the same number of points, then the curves must be identical.

Exercises.

1. Verify that if the limit in (10.2) exists for φ in a dense subset of $C_0^+(\mathbf{R}^n)$, then it exists for all $\varphi \in C_0^+(\mathbf{R}^n)$.
2. Let $E = C(\lambda) \times C(\lambda)$ with $0 < \lambda < 1/2$, recall 4.10. Show that $\dim(E \cap L) < \dim E$ for all lines L in $\mathbf{R}^2$. A generalization to self-similar sets and C^1 submanifolds is given in Mattila [5].
3. Prove that if Γ is a rectifiable curve in $\mathbf{R}^n$, then at $\mathcal{H}^1$ almost every point $x \in \Gamma$, the set $\Gamma \cap (L+x)$ is finite for $\gamma_{2,1}$ almost all $L \in G(n, 1)$.